



2025

第三届全国统计与数据科学联合会议

THE 3rd JOINT CONFERENCE ON STATISTICS AND DATA SCIENCE IN CHINA

A Causal Inference Task Based on Mendelian Randomization and Its Mediation Effect Analysis

Dier Tang, The University of Hong Kong / Jilin University

Joint work with

Prof. Shishun Zhao, Jilin University



Contents

- ◆ Background
- ◆ Materials and Methods
- ◆ Results
- ◆ Analysis

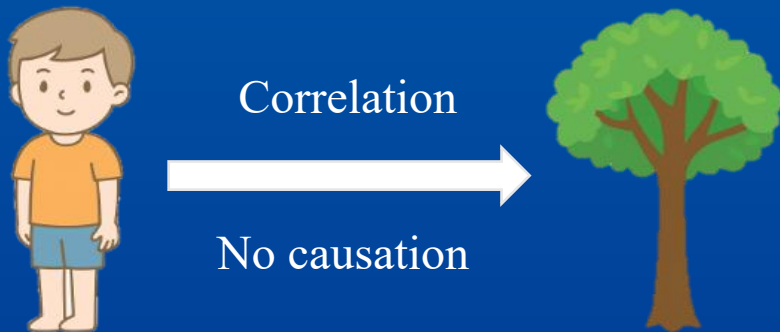


Background

- ◆ **Confounding** and **reverse causation** are difficulties in observational research.

Example 1: Confoundings

- I planted a tree while young.
- I grew older. Tree grew taller.
- My height statistically positively correlated with the height of tree.
- But there is **no causation**.
- **Time** is the confounding.

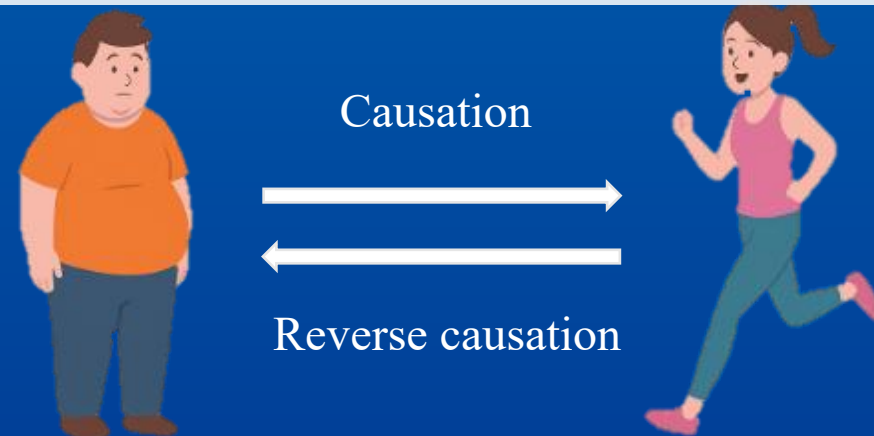


Example 2: Reverse causation

- According to observation, the fatter people usually exercise less.
- Is **less of exercise** → **fat**
or **fat** → **hard to do more exercise**

“Observed Correlation” unequal to “Causation”

direction is to tell which is the cause and the result.

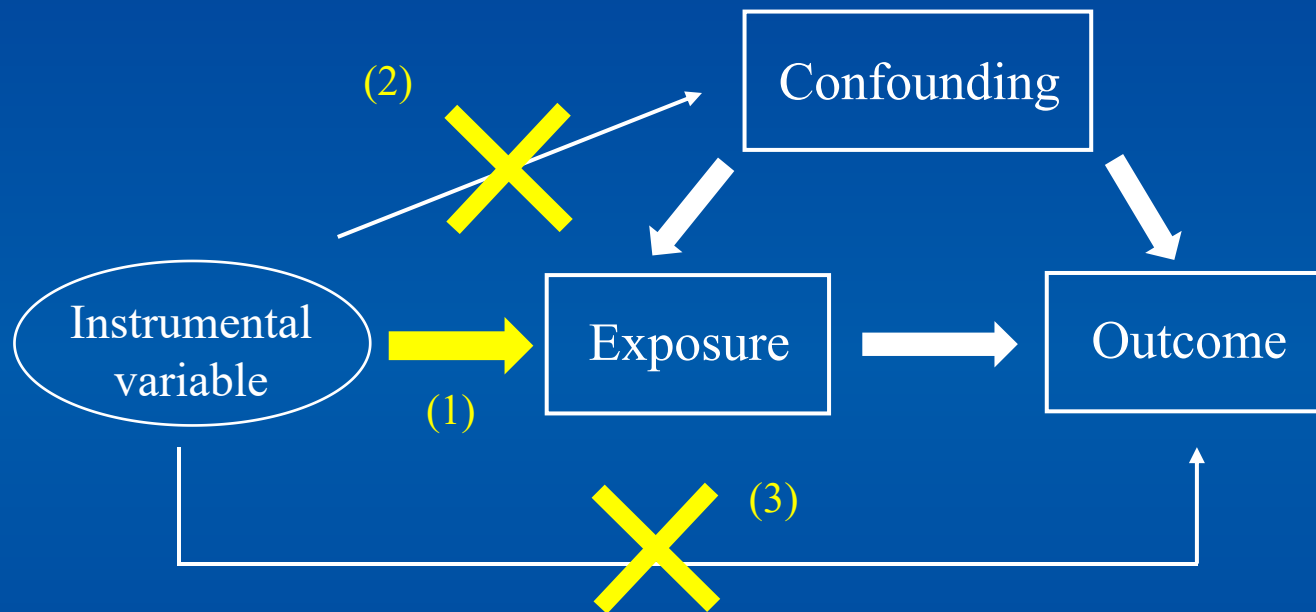




Background

- ◆ **Confounding** and **reverse causation** are difficulties in observational research.
- ◆ **Randomized controlled trial (RCT)** can effectively deal with them.
 - The known and unknown confounding factors are evenly distributed between the experimental group and the control group via random grouping, thus eliminating the influence of confounding.
 - The intervention takes place before and the outcome after (time sequence is clear), thus reducing the risk of reverse causality.
 - However, RCT usually has high cost, long time consumption, and may involve ethical issues, which makes it **difficult to achieve**.
- ◆ **Mendelian Randomization (MR)** was designed.
 - 1986, **Martijn Katan** first put forward the idea to use genetic variables as instrument variable.
 - 2003, **George Davey Smith** and **Shah Ebrahim** named it and established the theory.

Mendelian Randomization (MR)

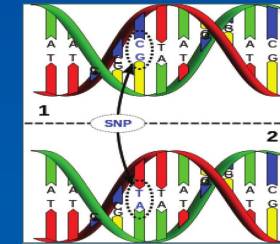


(1) Relevance Assumption: IVs **highly relevant** to exposure.

(2) Independence Assumption: IVs **independent** with any confoundings.

(3) Exclusion Restriction Assumption: IVs can **only** affect the outcome through exposure, **not directly** relevant to the outcome.

◆ MR takes **single nucleotide polymorphism (SNP)** as IVs.



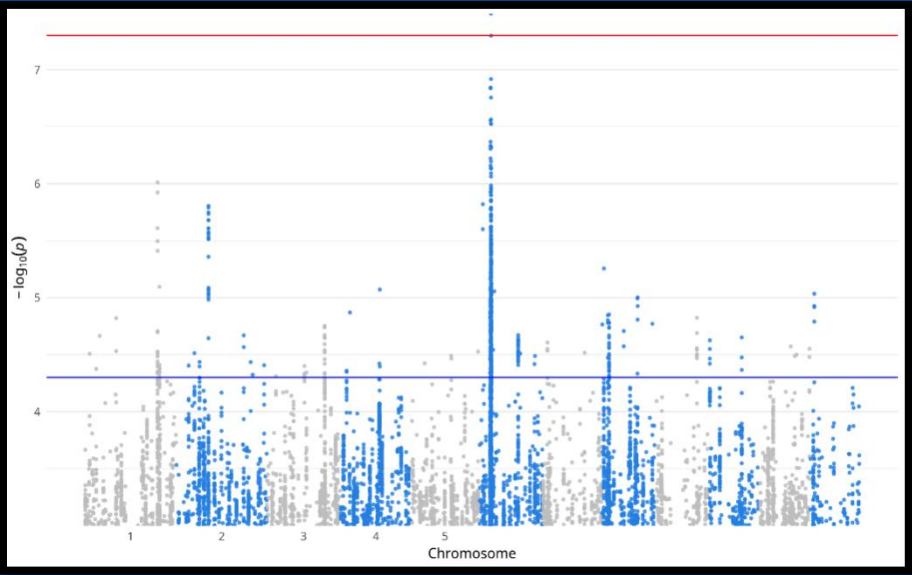
- Setting thresholds to satisfy assumption (1)
- Carrying out tests such as horizontal pleiotropic test to ensure assumption (3)
- According to Mendel's Law of Independent Assortment, genes for different traits assort independently of one another during gamete formation, which meets assumption (2)



Data Source

Genome-wide association studies (GWAS)

- GWAS summary data
- Open and free
- <https://gwas.mrcieu.ac.uk>



Exposure data (from UK Biobank):

Exposure	Sample size	nSNP	Year
Use of Sun/UV Protection	459416	9851867	2018
Ease of Skin Tanning	453065	9851867	2018

Outcome data:

- 3 types of **skin cancer** (Basal cell carcinoma, Malignant lymphoma, Malignant melanoma)
- 18 types of **non-skin cancer** (Leukemia, Brain tumor, Gastric cancer, Thyroid cancer and so on)

Mediator data:

- Based on previous studies, 45 kinds of mediating factors were included in this study, including inflammatory factor such as IL-10, enzyme such as MMP-12, and vitamin such as Vitamin D.



Methods

◆ Tool for research

- Software R (version 4.2.1)
- Using R package “TwoSampleMR” and “ieugwasr”

◆ Select IVs from exposure data:

- Setting $p < 1e-8$ as the threshold.
- Remove Linkage disequilibrium: set $r^2=0.001$, genetic distance=10000kb.
- Remove weak IVs: Use F statistics to detect weak IVs. Remove those whose $F < 10$.

◆ Use 5 MR methods to make causal inference:

- Inverse variance weighted (IVW): most efficient, but sensitive to pleiotropy.
- MR Egger: can deal with pleiotropy, but sensitive to outliers and violations of InSIDE assumption.
- Weighted median: robust to outliers, sensitive to addition/removal of genetic variants.
- Simple mode: robust to outliers, but tend to underestimate the real effect.
- Weighted mode: same to “Simple mode”.

◆ Judgement of significant results

- The result of IVW is significant.
- Besides IVW, at least 1 method present significant result.
- For all significant results, the values of beta have the same sign (the directions of OR are the same).



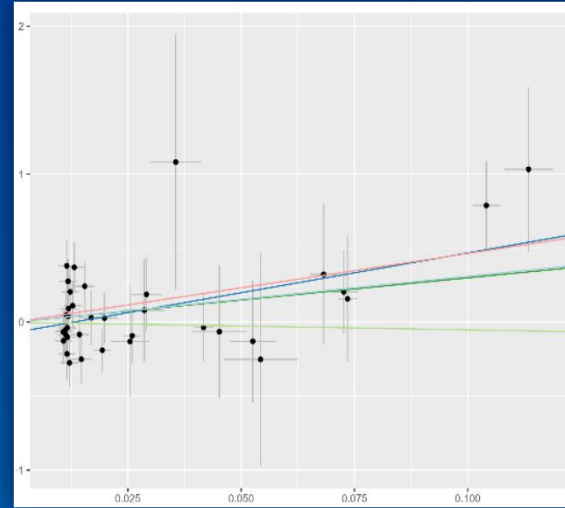
Materials and Methods

◆ Sensitivity analysis

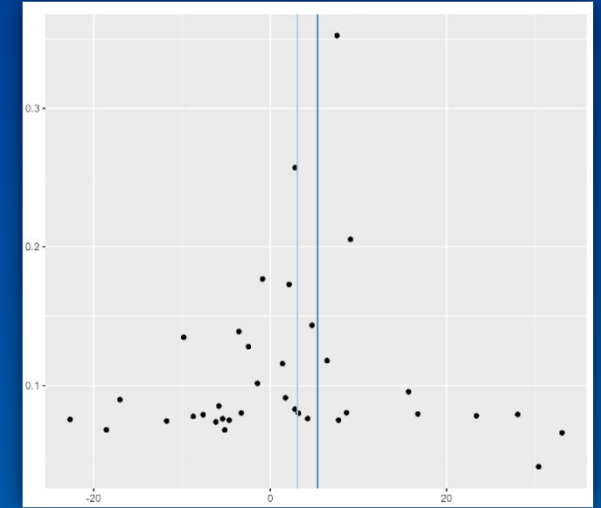
- Heterogeneity test: **Cochran's Q test**
- Horizontal pleiotropic test: **MR Egger intercept test**

◆ Results visualization

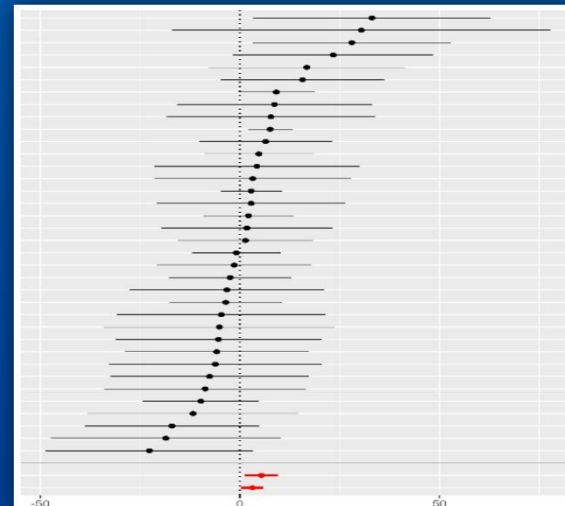
- Scatter plot: to see each SNP's effect on exposure and outcome.
- Funnel plot: symmetrical plot reveals relatively good results.
- Forest plot: to see each SNP's single causal effect.
- Leave-one-out plot: to see the effect while remove each SNP.



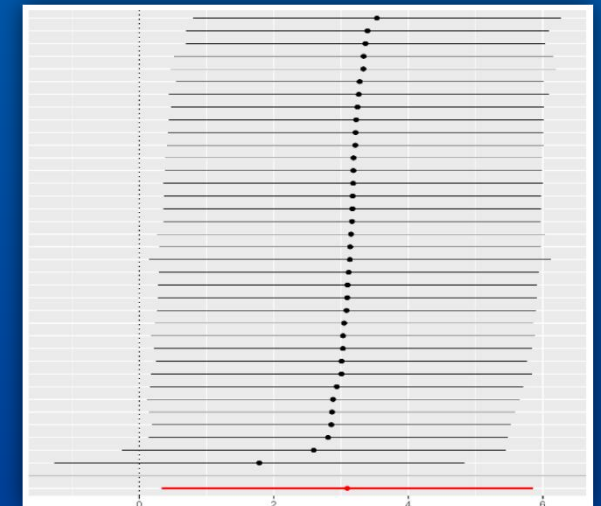
Scatter plot



Funnel plot



Forest plot



Leave-one-out plot

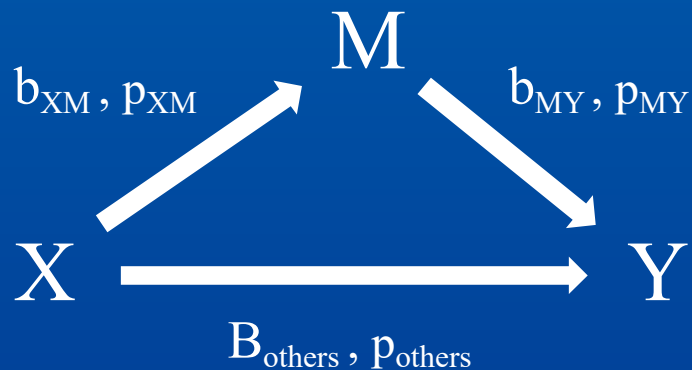


◆ Mediation effect analysis

- MR-based mediation effect analysis [C. Relton and G.D. Smith, 2012]
- Using IVW method in each analysis
- Total effect (X: Exposure, Y: Outcome):



- Mediation effect of mediator M:



- For each mediator, if $p_{XY} < 0.05$, $p_{XM} < 0.05$, $p_{MY} < 0.05$, then:

(1) Estimate total causal effect value b_{XY}

(2) Estimate the causal effect of exposure on mediator b_{XM} , and the causal effect of mediator on outcome b_{MY}

(3) Calculate mediation effect value:

$$b_{\text{indirect}} = b_{XM} \cdot b_{MY}$$

$$b_{\text{others}} = b_{XY} - b_{\text{indirect}}$$

(4) Calculate mediator proportion:

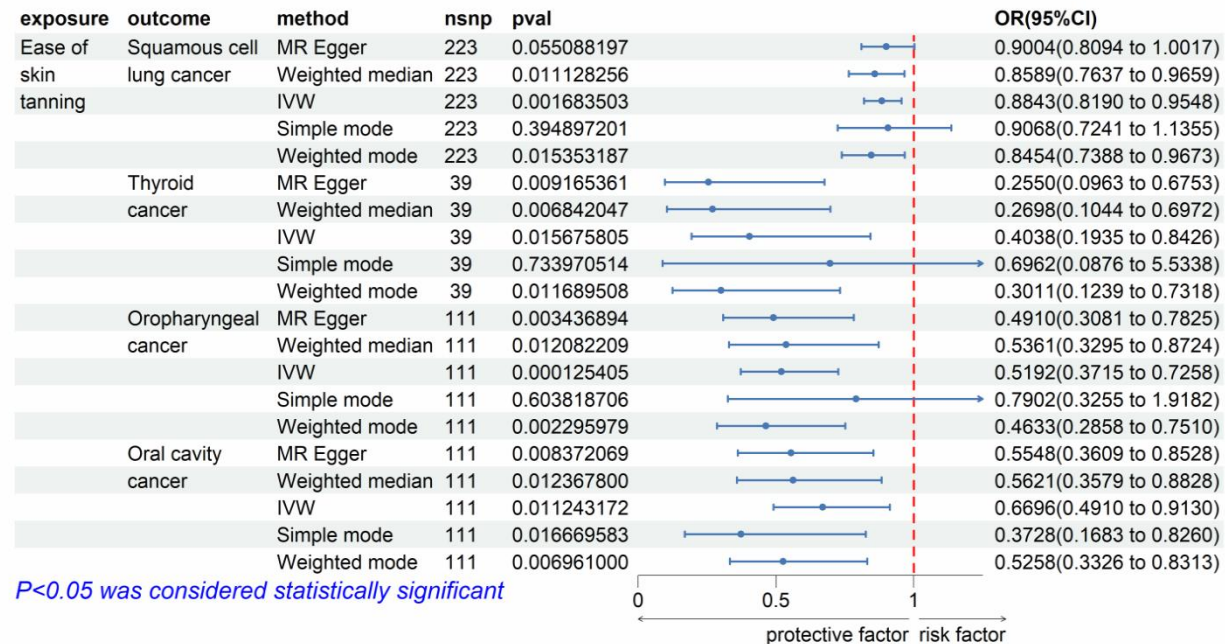
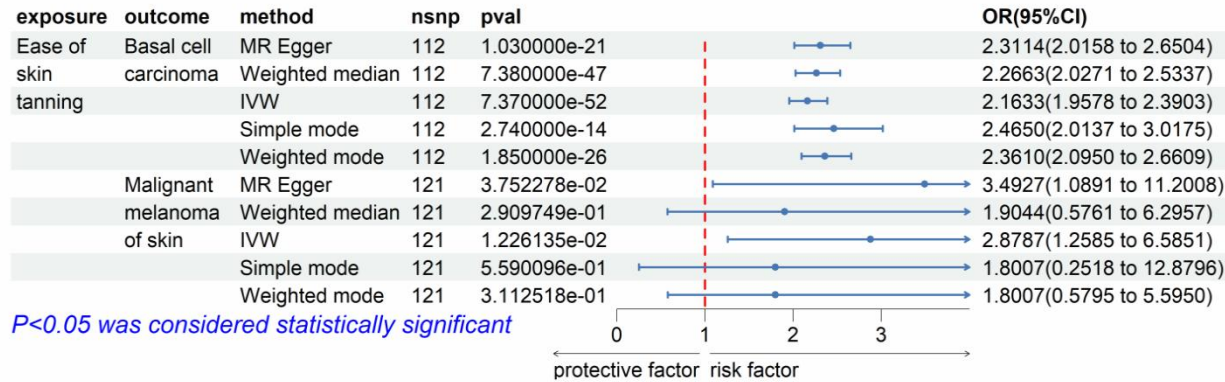
$$R_M = b_{\text{indirect}} / b_{XY}$$

◆ Causal inference and sensitivity analysis

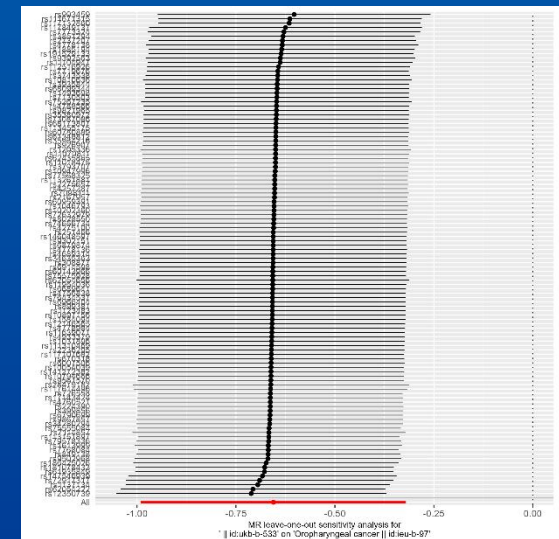
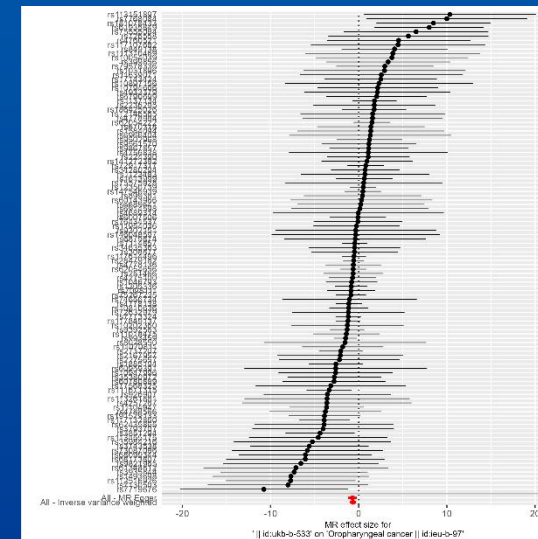
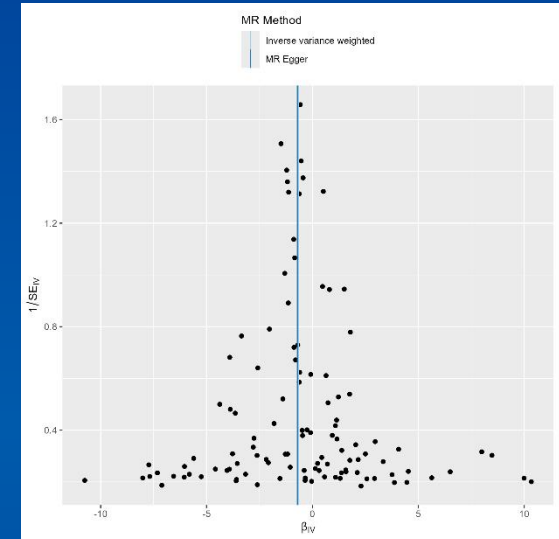
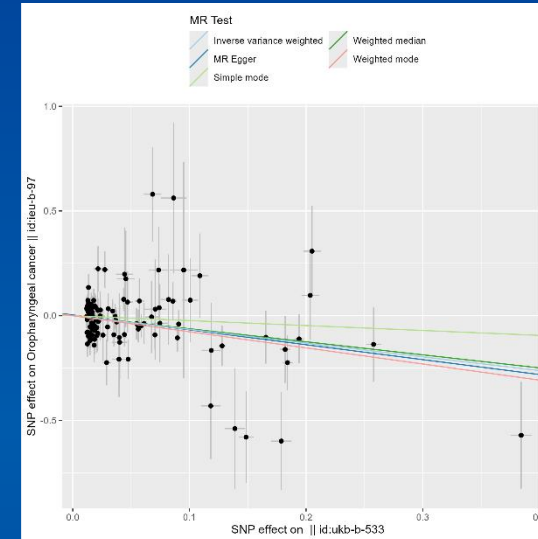
Exposure	Outcome	Causation	pval of Cochran's Q test	pval of MR Egger intercept test
Use of Sun/UV Protection	Basal cell carcinoma (skin)	positive	0. 009	0. 032
	Malignant melanoma (skin)	positive	0. 325	0. 160
	Squamous cell lung cancer (non-skin)	negative	0. 135	0. 511
	Thyroid cancer (non-skin)	negative	0. 324	0. 295
	Oropharyngeal cancer (non-skin)	negative	0. 056	0. 542
Ease of Skin Tanning	Basal cell carcinoma (skin)	positive	0. 013	0. 170
	Malignant melanoma (skin)	positive	0. 257	0. 526
	Squamous cell lung cancer (non-skin)	negative	0. 292	0. 633
	Thyroid cancer (non-skin)	negative	0. 713	0. 167
	Oropharyngeal cancer (non-skin)	negative	0. 190	0. 735
	Oral cancer (non-skin)	negative	0. 416	0. 220

Results

Forest plot of ease of skin tanning



“Ease of skin tanning – Oropharyngeal cancer”



Results

◆ Further processing and explanation

- Heterogeneity, no horizontal pleiotropy:
The results of **IVW** and **Weighted median** are considered reliable. [Burgess et al., 2013]
- Horizontal pleiotropy:
MR Egger have higher priority. [Bowden, 2015]
- Both for “Use of Sun/UV Protection - Basal cell carcinoma” and “Ease of Skin Tanning - Basal cell carcinoma”, **IVW**, **MR Egger** and **Weighted median** methods all present significant results. **InSIDE assumption holds**. [Burgess, 2017]

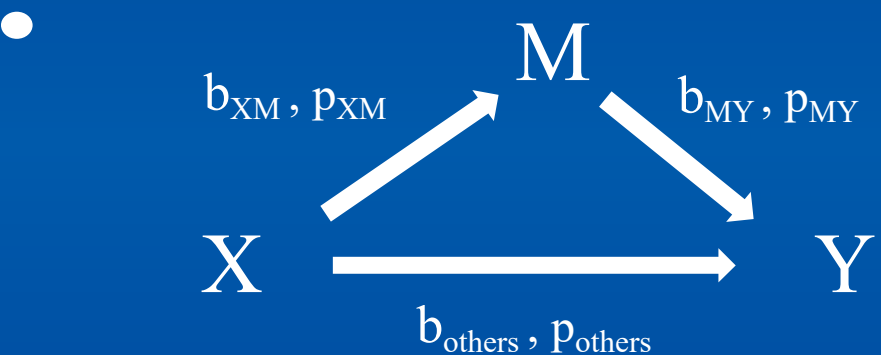
exposure	outcome	method	nsnp	pval	
Use of sun/uv protection	Basal cell carcinoma	MR Egger	32	1.850000e-09	
		Weighted median	32	7.300000e-41	
		IVW	32	9.300000e-21	
		Simple mode	32	3.560000e-01	
		Weighted mode	32	4.610000e-15	
exposure	outcome	method	nsnp	pval	
Ease of skin tanning	Basal cell carcinoma	MR Egger	112	1.030000e-21	
		Weighted median	112	7.380000e-47	
		IVW	112	7.370000e-52	
		Simple mode	112	2.740000e-14	
		Weighted mode	112	1.850000e-26	

- The results of this study are **reliable**.

◆ Mediation effect analysis

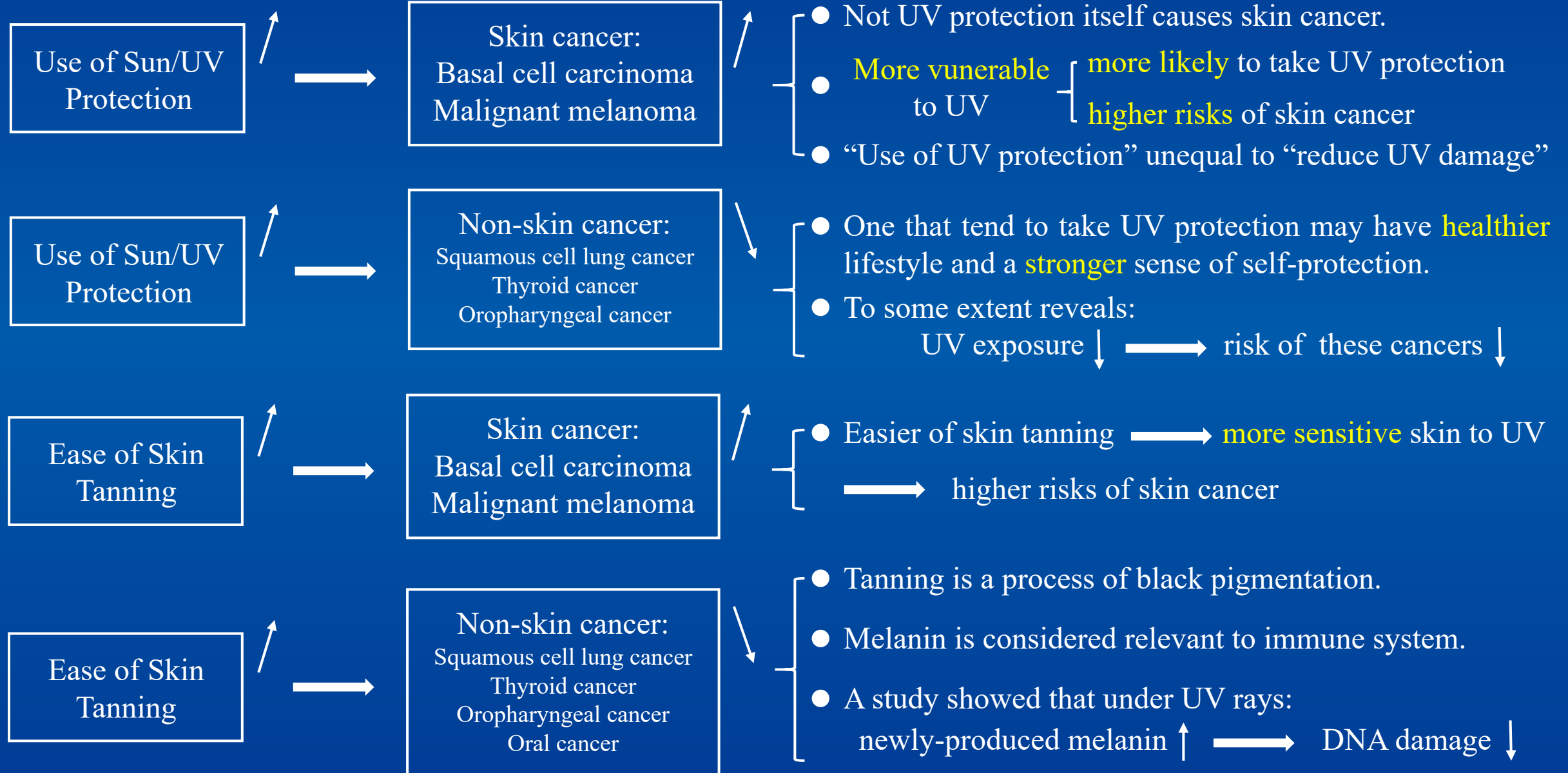
- X: Use of Sun/UV Protection, Y: Oropharyngeal cancer

$$X \xrightarrow{b_{XY}, p_{XY}} Y \quad (b_{XY} = -1.200, p_{XY} = 0.041)$$



Mediator	b_{XM}	p_{XM}	b_{MY}	p_{MY}	mediator proportion
IL-6	0.183	< 0.05	-0.206	< 0.05	3.1416%
TNFSF14	0.150	< 0.05	-0.197	< 0.05	2.4625%
TGF-b	0.162	< 0.05	-0.092	< 0.05	1.2428%

■ Analysis



Thanks for your listening!

Dier Tang

The University of Hong Kong / Jilin University

